There can be multiple objectives for ranking the news items. One is to maximise the engagement of the user with the news items. Another can be to rank the items based on their information quality. Of course, you will also like to filter out news items that may be spam, misinformation or clickbait. Given these multiple objectives, can you explain how you will build a ranking system for personalised news feeds? What will your design choices be?

6. Continuing from (5) above, for problems that have multiple objectives, will you prefer to build a single ML model? Or would it be better to build separate models, each optimising for a separate objective and then combine their individual scores/ranks into an aggregate score? What are the advantages and disadvantages of using a single model versus multiple models?
7. Continuing from (6) above, how will you collect data to train your models? And how will you handle the cold start problem, where you initially don't have data for most users?
8. One of the shortcomings of existing supervised learning techniques is that they need a large number of examples in their training data in order to perform a task well. This is unlike human beings, who can learn a new task with very few labelled examples. Do you know of ML techniques that can work around the need for a large number of labelled examples to train a model?
9. Human beings have the inherent capability to extrapolate their knowledge from one task to other related tasks. For example, if you know how to ride an ordinary bicycle, it will be quite easy for you to ride a two-wheeler or a motor bicycle. ML models, however, are not that good in transferring their knowledge from one task to another. Consider this problem as an example. You have a model that has been built for emotion classification in text. Given a piece of text, the model can label the text as one of several emotion classes, namely, a multi-class classification problem. Let us assume that our input is a piece of text in English and the output is a label mapping to one of eight emotion classes. We want to use this model for the related problem of sentiment analysis. Here we have three output classes, namely, positive, negative and neutral. Which one is the sentiment associated with the given piece of text? Can you explain how you will use transfer learning techniques to solve this problem? What are the advantages and disadvantages of using a transfer learning based approach as opposed to building a dedicated sentiment analysis model from scratch?
10. You are given the problem of building a sentiment analysis model (as in the above problem). You have a data set of 100,000 samples. However, the data set is yet to be labelled. It will cost considerably to label all the 100,000 samples. Given your budget, you can only get 30,000 samples labelled. How will you decide which of the 30,000 samples you should get annotated? Will you select 30,000 samples randomly from your given data? Or will you deploy any other heuristic to select the samples that need to be labelled? Do you know how active learning techniques work?
11. Once you have built and deployed an ML model for your task, how often will you need to retrain and update the model? What are the criteria you will use to decide when retraining is required? Also, how will you collect data during deployment to enable retraining?
12. You are given the problem of identifying anomalous transactions in bank transactions data. You need to build an ML model that can recognise such fraudulent transactions. However, the data set is highly imbalanced since fraudulent transactions are very rare in the real world. You have only 0.5 per cent of fraudulent transactions and the rest are normal transactions. Will you use sampling techniques to handle the imbalanced data problem? Will your choice change if you are using a classical ML model versus a deep learning neural network system? Can you explain the rationale behind your choice?
13. You have been asked to build an ML model that can identify toxic/hate speech content in a user's feed on a content sharing platform. You have built two models – one model has more false positives and another model has more false negatives. Which model will you go with and why?
14. Continuing from (13) above, now you have two models – model A has a F1 score of 90 per cent but has a longer inference latency, while model B has a F1 score of 83 per cent but has a lower inference latency. Which model will you choose and why? Now, instead of a content sharing platform, you have been asked to use these models to scrub emails of objectionable content. Will your choice of model change for this use case? If so, why?
15. Continuing from question (13) above, instead of choosing between model A and model B, will you consider making an ensemble of the two models for your task? If not, why?

Feel free to reach out to me over LinkedIn/email if you need any help in your coding interview preparation. If you have any favourite programming questions/ software topics that you would like to discuss on this forum, please send them to me, along with your solutions and feedback, at *sandyasm_AT_yahoo_DOT_com*. Stay healthy and stay safe. END

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